

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 10%

QUESTIONS: 15

Total Marks:20

Annexure A

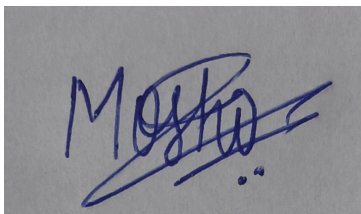
Comparative Analysis

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Moshika M', is written over a grey rectangular background.

Annexure A

Short Answer Test

1. Compare K-Nearest Neighbour and Case-Based Learning using a real-world example such as medical diagnosis. Analyze differences in memory usage, similarity measures, and prediction strategy.

ANSWER:

Consider a medical diagnosis system where a new patient's symptoms, age, blood pressure, test results, and other medical features are compared with previously diagnosed patients.

K-Nearest Neighbour

KNN stores training examples and classifies a new patient by finding the K most similar patients. The predicted diagnosis is usually the majority class among these neighbours.

Example:

If a new patient is similar to 5 previous patients and 4 of them were diagnosed with diabetes while 1 was not, KNN predicts diabetes.

Case-Based Learning

Case-Based Learning stores complete past cases and their solutions. When a new patient arrives, the system retrieves the most relevant previous case(s), adapts the previous solution if necessary, and uses it to make a diagnosis.

Example:

A new patient with symptoms similar to a previous patient diagnosed with pneumonia may be diagnosed similarly, after considering differences in symptoms and test results.

Comparison

Feature	K-Nearest Neighbour	Case-Based Learning
Basic idea	Classifies using K closest training points	Retrieves and adapts previous cases
Memory usage	High because training instances are stored	High because complete cases are stored
Similarity measure	Usually mathematical distance such as Euclidean or Manhattan distance	Can use domain-specific similarity measures

Prediction strategy	Majority voting or distance-weighted voting	Retrieve → reuse → adapt → retain
Adaptation	Limited	Strong; previous solutions can be modified
Training	Very little explicit training	Requires maintaining and organizing cases
Medical use	Disease classification	Diagnosis and treatment based on similar historical cases

Conclusion

KNN is simpler and works well when numerical features can be represented using a meaningful distance measure. Case-Based Learning is more suitable when previous medical cases contain rich contextual information and solutions need to be adapted. KNN focuses mainly on neighbour voting, whereas CBL focuses on reusing and adapting past experiences.

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

ANSWER:

Consider a dataset representing the relationship between temperature and electricity consumption. The objective is to approximate the underlying function and predict electricity consumption for a new temperature value.

Locally Weighted Regression

LWR gives more importance to training points that are close to the query point and less importance to distant points.

For a query point x_q , a common weight function is:

$$w_i = e^{-\frac{(x_q - x_i)^2}{2\tau^2}}$$

where τ controls the size of the local region.

Thus, LWR builds a local model for every query.

Radial Basis Function Network

An RBF network uses radial basis functions, commonly Gaussian functions, as hidden-layer neurons:

$$\phi_i(x) = e^{-\frac{\|x - c_i\|^2}{2\sigma_i^2}}$$

where c_i is the centre of the RBF neuron and σ_i controls its spread.

The outputs of these basis functions are combined to produce the final prediction.

Comparison

Feature	LWR	RBF Network
Locality	Uses nearby training points for each prediction	Uses basis functions centred at selected locations
Model construction	Local model constructed at query time	Global network learned during training
Computational cost	High during prediction	Higher during training, generally faster prediction
Memory	Usually stores training data	Can use a smaller set of centres
Smoothness	Depends strongly on bandwidth	Usually provides smooth predictions
Adaptability	Very high; directly uses new observations	Requires retraining/update for significant changes
Prediction	Local weighted regression	Weighted combination of RBF activations

Example

Suppose the dataset contains:

Temperature (°C)	Electricity Usage (kWh)
20	100
25	120
30	160
35	220
40	300

For a prediction at **31°C**, LWR gives high weights to the observations at 30°C and 35°C while giving much lower weights to 20°C and 25°C.

An RBF network may have centres around 25°C, 30°C, 35°C and 40°C. The prediction at 31°C is obtained from the combined activation of these nearby RBF units.

Conclusion

LWR is highly local and adaptive, but repeatedly calculating a local model makes prediction computationally expensive. RBF networks perform more computation during training but provide fast and smooth predictions after training. Therefore, LWR is useful for datasets that change frequently, while RBF networks are preferable when repeated fast predictions are required.

3. Analyze the differences between K-Nearest Neighbour and Radial Basis Function models in classification problems. Provide an example and compare decision boundaries, training time, and generalization ability.

ANSWER:

Consider a handwritten digit classification problem where features extracted from digit images are used to classify digits from 0–9.

KNN Classification

KNN stores the training examples. For a new digit, it calculates the distance between the new sample and training samples, selects the K nearest samples, and assigns the class based on majority voting.

For example, if K=5:

- 3 nearest samples → Digit 3
- 1 nearest sample → Digit 8
- 1 nearest sample → Digit 5

The predicted class is **Digit 3**.

RBF Classification

An RBF classifier transforms input data through radial basis functions. Each RBF unit responds strongly to inputs near its centre.

The output can be represented as:

$$y_k(x) = \sum_{j=1}^m w_{kj} \phi_j(x)$$

The class with the highest output is selected.

Comparison

Feature	KNN	RBF Model
Training time	Very low	Higher
Prediction time	High for large datasets	Generally low
Memory usage	High	Usually lower
Decision boundary	Can be irregular and highly local	Smooth and nonlinear
Parameter	K and distance metric	Centres, widths and weights
Generalization	Depends strongly on K and data density	Usually good with properly selected centres
Adaptability	Easy to add new training examples	Requires model updating/retraining

Scalability	Prediction becomes expensive with large datasets	Better for repeated predictions
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Decision Boundaries

KNN can produce complex and irregular decision boundaries because the boundary depends directly on individual training points.

RBF models produce smoother boundaries because nearby points activate RBF units, and the final decision is based on the combined response of these units.

Example

Suppose two classes represent:

- **Class A:** healthy patients
- **Class B:** patients with a particular disease

KNN determines the class of a new patient based on nearby historical patients. RBF learns representative centres for regions corresponding to the two classes and creates a nonlinear boundary between them.

Conclusion

KNN is easier to implement and requires almost no explicit training, making it useful when the dataset changes frequently. However, prediction can become slow for very large datasets. RBF networks require more training but generally provide faster prediction and smoother decision boundaries, making them useful for large-scale classification applications.

4. Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

ANSWER:

Consider an online movie recommendation system. The system needs to recommend movies to a user based on previous users' preferences and ratings.

Case-Based Learning

CBL stores previous recommendation cases. A case might contain:

- User preferences
- Previously watched movies
- Ratings
- Genres
- Previous recommendations
- User response to recommendations

When a new user or situation appears, the system retrieves similar cases and adapts their recommendations.

Example:

A user who enjoys science-fiction, action, and thriller movies may be matched with previous users having similar preferences. Their successful recommendations can then be adapted for the new user.

Locally Weighted Regression

LWR can predict a numerical value, such as the rating a user might give to a movie.

Similar users or movies receive higher weights, while less similar ones receive lower weights. The predicted rating can then be used to rank movies.

Comparison

Feature	Case-Based Learning	Locally Weighted Regression
Main purpose	Reuse previous cases	Predict numerical values
New data	Retrieves similar cases and adapts them	Uses nearby observations to build a local model
Adaptability	Very high	Very high
Memory usage	Stores complete cases	Usually stores training observations
Computational efficiency	Retrieval can become expensive with many cases	Prediction can be expensive because local regression is performed
Output	Previous solution/recommendation adapted to user	Numerical prediction such as rating
Learning style	Experience-based	Local statistical learning
Best suited for	Context-rich recommendations	Rating/score prediction

Handling New Data

CBL:

When a new user appears, the system searches for similar previous users and reuses their successful recommendation patterns. The new experience can also be stored as a new case.

LWR:

When a new user/movie combination appears, LWR identifies nearby data points and calculates a prediction using their weighted influence.

Computational Efficiency

CBL may require searching through a large case database. Efficient indexing and similarity retrieval can reduce this cost.

LWR can also be computationally expensive because a local regression calculation may be performed for every new query.

Conclusion

Case-Based Learning is better when recommendations depend on rich previous experiences and contextual information. LWR is more appropriate when the goal is to predict a numerical rating or preference score from nearby examples. CBL emphasizes retrieving and adapting experiences, whereas LWR emphasizes local mathematical prediction.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	15	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	20	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand